Python for Data Analysis: glossary

Every word in the course, with the session where it first appears. This list is generated from the "Words for today" cards, so a word cannot be taught without reaching this handout.

Word	What it means	Think of it like this	Session
and, or	and needs both conditions True. or needs at least one.	Entry needs a ticket and an ID. A discount needs a student card or a coupon.	DA04
argument	The actual value you give a function when you call it.	The answer you write in the blank.	DA06
array	A row of numbers that NumPy can calculate with all at once.	A strip of numbers on a ruler.	DA13
assert	Stop with an error if a statement is False. Say nothing if it is True.	A teacher checking the answer without saying well done.	DA06
axis label	The words along an axis that name the quantity and its unit.	The heading of a column.	DA11
bar chart	Bars compare amounts between categories.	Piles of stock side by side.	DA11
Boolean	A value that is either True or False.	A light that is on or off.	DA04
box plot	A box from the lower to the upper quartile with a line at the median and dots for outliers.	A five-number summary drawn as a picture.	DA14
caption	One sentence under a chart that says what it shows.	The words under a photograph.	DA21
cause	One thing makes another happen. A correlation alone does not show it.	Rain makes the ground wet. Wet ground does not make rain.	DA15
cell	One box in a notebook. A text cell explains. A code cell runs.	One paragraph on a page of the exercise book.	DA01
chart report	A short notebook of charts, numbers and sentences that answers the project questions.	A news page with pictures and captions.	DA21
cleaning log	A record of each problem found and what was done about it.	The tailor's note of every change to a uniform.	DA10
column	One kind of information in a table, such as units.	One vertical strip of the register.	DA07
comment	A note for people that starts with # and is ignored by Python.	A pencil note in the margin.	DA03
comparison	A question that gives True or False, such as units >= 5.	Asking whether a pupil is tall enough to ride.	DA04
copy	A separate table to work on, so the raw table stays untouched.	A photocopy of the receipt book.	DA10
correlation	A number from -1 to 1 for how closely two columns move together.	Taller pupils usually have longer feet.	DA15

Word	What it means	Think of it like this	Session
CSV	Comma-separated values: a plain-text file that holds a table.	A register copied by hand with commas between the boxes.	DA07
data dictionary	A list of every column with its meaning and unit.	The key to a map.	DA19
data source note	A line that says where the data came from and its limits.	The credit under a map.	DA21
data type	How a column is stored: whole number, decimal or text.	Whether a box holds numbers or words.	DA10
DataFrame	The table object in pandas, with rows and named columns.	The register after it has been opened in Python.	DA07
denominator	The whole that a proportion is measured against.	The 20 in 12 out of 20.	DA09
dictionary	Named pairs inside braces. A key finds its value.	A price board: the name of the item gives its price.	DA05
disclosure	A plain statement of any help or AI tool you used.	Writing your sources at the end of an essay.	DA19
distribution	How the values of one column are spread out.	How the pupils spread across height bands.	DA14
duplicate	A row that repeats another row exactly.	The same sale written twice.	DA10
edge case	A value at the boundary where a rule might break, such as zero.	A customer who buys nothing.	DA06
element-wise	Doing the same calculation to every number in an array at once.	Adding 100 shillings to every price on the board.	DA13
else	Run these lines when the condition is False.	Otherwise, go to the queue.	DA04
error message	Python explaining why it stopped. Read the last line first.	A teacher's comment on what went wrong.	DA03
exclude	Leave a row out and count it, because it cannot be repaired.	Setting aside the damaged goods and counting them.	DA16
exploratory analysis	Looking at the data from several sides to see what is there before you answer a question.	Walking round a shop before you count the stock.	DA14
expression	A calculation Python can work out, such as 3 * 2000.	One line of working in a maths exercise.	DA03
fan-out	A merge that makes extra rows because a key appears twice, and inflates every total.	One receipt copied for every duplicated price line.	DA17
file	A saved piece of work with a name, such as camp_shop.csv.	A page stored in a folder.	DA01
filter	Keep only the rows where a comparison is True.	Picking out the sales of 5 units or more.	DA08
float	A number with a decimal part.	A price of 2500.5 shillings.	DA02

Word	What it means	Think of it like this	Session
function	A named set of instructions you can use again.	A recipe with a name.	DA06
group	Gather rows with the same label before you calculate.	Sorting the sales into piles by item.	DA09
group comparison	Drawing the same chart for each group side by side.	Comparing the three shops on one page.	DA14
header	The name written at the top of a column.	The title of a column in the register.	DA07
heatmap	A table of numbers coloured by size.	A map shaded by rainfall.	DA15
histogram	Bars show how many values fall in each range.	Counting pupils in each height band.	DA11
hue	A colour that shows which group each dot belongs to.	A different coloured pen for each shop.	DA15
if	Run the indented lines only when the condition is True.	Enter only if you have a ticket.	DA04
indentation	Four spaces at the start of a line show which lines belong to the if.	The inside of a bracket in a sentence.	DA04
index	The position of a value in a list. Counting starts at 0.	The numbered seats in a bus, with the driver as seat 0.	DA05
integer	A whole number with no decimal part.	Counting 3 books.	DA02
key	The name used to look up a value in a dictionary.	The name on a price board.	DA05
left merge	Keep every row of the first table, even when nothing matches.	Keep every receipt, even for a missing price board line.	DA17
library	Code written by other people that you load with import.	A toolbox you borrow.	DA07
licence	The rules that say how you may use a dataset.	The terms on a borrowed book.	DA19
limitation	What the data or method cannot tell us.	The part of the map that is blank.	DA19
line chart	A line shows how one thing changes over ordered dates.	A temperature record.	DA11
line number	The place in the cell where Python found the problem.	The page and line in a book.	DA03
list	Several values in order inside square brackets.	A shopping list.	DA05
lookup	A dictionary that turns one value into another.	The price board.	DA16
loop	Repeat the indented lines once for each value.	Counting stock shelf by shelf.	DA05

Word	What it means	Think of it like this	Session
mask	A column of True and False values, one for each row, that a filter uses.	A ticked list beside the register.	DA08
matplotlib	The library that draws charts. We load it as plt.	A pen and a ruler for drawing.	DA11
mean	The total shared out equally: add the values and divide by how many.	Sharing a bag of mandazi equally.	DA09
median	The middle value once the values are in order.	The pupil standing in the middle of a line sorted by height.	DA09
merge	Join two tables by matching their key columns.	Matching each receipt with its line on the price board.	DA17
merge key	A column that identifies a row and is shared by two tables, such as product_id.	The item number on both the price board and the receipt.	DA17
missing value	A cell with nothing recorded. It is not the same as zero.	A blank in the register.	DA10
new column	A column worked out from other columns and added to the table.	Multiplying units by price on every line.	DA08
normalise	Make the same thing look the same everywhere, such as one spelling and one case.	Writing every name with a capital first letter.	DA16
notebook	A page where you write instructions for the computer and see the answers under them.	An exercise book that answers back.	DA01
NumPy	The library of fast number tools that pandas is built on. We load it as np.	The calculator inside the register.	DA13
operator	A symbol that calculates: + - * / and **.	The buttons on a calculator.	DA03
order of operations	Brackets first, then powers, then multiply and divide, then add and subtract.	The rule that makes 2 + 3 * 4 equal 14.	DA03
outlier	A value far from the rest.	One huge order in a week of small sales.	DA13
output	What a cell shows after it runs.	The answer written under the question.	DA01
parameter	The name for an input inside a function.	The blank in a form.	DA06
parse	Read text and turn it into a proper date or number.	Reading a handwritten date and writing it as a real date.	DA16
percentile	The value below which a given share of the numbers fall.	The mark that 75 out of 100 pupils did not beat.	DA13
pipeline	A series of cleaning steps that always run in the same order.	A production line.	DA16
project question	One question a table can answer with numbers.	Which item sells most, asked of a till roll.	DA19
proportion	A part divided by the whole. Times 100 it becomes a percentage.	12 pupils out of 20 passed.	DA09

Word	What it means	Think of it like this	Session
raw data	The file exactly as it arrived. You never overwrite it.	The original receipt book.	DA10
reconcile	Check that two routes to the same total agree, or explain why not.	Counting the cash tin and the receipts and comparing them.	DA17
residual	The difference that is left after you have done everything right.	The small change nobody can account for.	DA16
return	Send the result back to where the function was called.	The cashier handing you the change.	DA06
revenue	Money received from sales: units times price. It is not profit.	The takings before paying for stock.	DA08
row	One record in a table, such as one sale.	One line in the shop register.	DA07
run	Tell the computer to carry out the code in a cell. Click the cell and press Shift and Enter.	Handing in a question and waiting for the answer.	DA01
running total	A number that grows as each value is added.	The cash tin during the day.	DA05
scatter plot	Dots show how two numbers go together.	Each pupil as a dot by height and age.	DA11
scope	How much the project covers, and what it leaves out.	The fence round a garden.	DA19
seaborn	A library that draws statistical charts. It sits on top of matplotlib and takes a DataFrame directly.	A set of ready-made stencils for charts.	DA14
select	Name a column in square brackets to look only at it.	Reading only the price strip of the register.	DA08
skewed	Bunched at one end with a long tail at the other.	Most pupils short and a few very tall.	DA14
sort	Put rows in order from smallest to largest or the other way.	Lining up pupils by height.	DA09
standard deviation	A number for how far values typically sit from their mean.	How far pupils usually stand from the middle of the line.	DA13
strength	How close the correlation is to 1 or -1. Near 0 means weak.	How tightly the dots hug a line.	DA15
string	Text inside quotation marks.	A name written on a label.	DA02
sum	All the values added together.	Adding up the day's takings.	DA08
table	Information arranged in rows and columns.	The sales page of a shop register.	DA01
type	The kind of value: integer, float or string.	Knowing whether a box holds pens or juice.	DA02
unique	Appears once only. A key must be unique in the table it looks up.	One item number per product.	DA17

Word	What it means	Think of it like this	Session
value	One piece of information, such as 2000 or Juice.	One entry on a receipt.	DA02
variable	A name that points to a value.	A label on a box of stock.	DA02